

Construction π_A over Multiquadratic Fields for Compound Block-Fading Wiretap Channels

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Definition 1 (Biquadratic Fields, (Jarvis 2014))

A biquadratic field is a number field that is the smallest Galois extension of the rational number field $\mathbb{Q}$ containing $\sqrt{a}$ and $\sqrt{b}$:

$$K = \mathbb{Q}(\sqrt{a}, \sqrt{b}) = \{x_0 + x_1\sqrt{a} + x_2\sqrt{b} + x_3\sqrt{k} \mid x_0, x_1, x_2, x_3 \in \mathbb{Q}\},$$

where a and b are distinct square-free integers and $k = ab/\gcd(a, b)^2$.

For a biquadratic field, $[K : \mathbb{Q}] = 4$, and K contains the three quadratic subfields $\mathbb{Q}(\sqrt{a})$, $\mathbb{Q}(\sqrt{b})$, and $\mathbb{Q}(\sqrt{k})$, where $k = ab/\gcd(a, b)^2$.

The ring of integers $\mathcal{O}_K$ of a biquadratic field $K = \mathbb{Q}(\sqrt{a}, \sqrt{b})$, admits an explicit integral basis depending on the congruence classes of a , b , and k modulo 4.

Prime splitting and CRT

For our lattice construction, we need to understand how rational primes factor in $\mathcal{O}_K$. In the Dedekind domain $\mathcal{O}_K$, $p\mathcal{O}_K = \prod_{i=1}^r \mathfrak{p}_i^{e_i}$, where the $\mathfrak{p}_i$ are distinct prime ideals.

We are mainly interested in the case where p **splits completely**:

$$p\mathcal{O}_K = \mathfrak{p}_1 \cdots \mathfrak{p}_n, \quad e_i = f_i = 1.$$

Proposition 1 (Marcus 1977)

If p splits completely in both $\mathbb{Q}(\sqrt{a})$ and $\mathbb{Q}(\sqrt{b})$, then it splits completely in $\mathbb{Q}(\sqrt{a}, \sqrt{b})$.

Chinese Remainder Theorem

If $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ are pairwise coprime ideals and $\mathfrak{P} = \prod_{j=1}^r \mathfrak{p}_j$, then

$$\mathcal{O}_K/\mathfrak{P} \cong (\mathcal{O}_K/\mathfrak{p}_1) \times \cdots \times (\mathcal{O}_K/\mathfrak{p}_r).$$

Main Example

Example: $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$, $p = 2$

Let $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$. Because $17 \equiv 33 \equiv 1 \pmod{8}$, the prime 2 splits in both $\mathbb{Q}(\sqrt{17})$ and $\mathbb{Q}(\sqrt{33})$. Hence 2 splits completely in K :

$$\langle 2 \rangle = \mathfrak{p}_1 \mathfrak{p}_2 \mathfrak{p}_3 \mathfrak{p}_4.$$

Using SageMath, we compute

$$\begin{aligned} \mathfrak{p}_1 &= \left\langle -\frac{23}{4} - \frac{7}{4}\sqrt{17} - \frac{5}{4}\sqrt{33} - \frac{1}{4}\sqrt{561} \right\rangle, \mathfrak{p}_2 = \left\langle -\frac{23}{4} - \frac{7}{4}\sqrt{17} + \frac{5}{4}\sqrt{33} + \frac{1}{4}\sqrt{561} \right\rangle, \\ \mathfrak{p}_3 &= \left\langle \frac{23}{4} - \frac{7}{4}\sqrt{17} + \frac{5}{4}\sqrt{33} - \frac{1}{4}\sqrt{561} \right\rangle, \mathfrak{p}_4 = \left\langle -\frac{27}{4} - \frac{5}{4}\sqrt{17} - \frac{3}{4}\sqrt{33} - \frac{1}{4}\sqrt{561} \right\rangle. \end{aligned}$$

Euclidean Lattice

A lattice in $\mathbb{R}^N$ is the set of all integer combinations of N linearly independent vectors $\mathbf{v}_1, \dots, \mathbf{v}_N$:

$$\Lambda = \left\{ \sum_{k=1}^N u_k \mathbf{v}_k \mid u_k \in \mathbb{Z} \right\} \subset \mathbb{R}^N.$$

The vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_N\}$ form a basis of Λ , and

$$G = [\mathbf{v}_1 \mid \cdots \mid \mathbf{v}_N]$$

is a generator matrix.

Algebraic Lattices and Canonical Embeddings

Let K be a number field of degree n . Its embeddings into $\mathbb{C}$ define the canonical embedding

$$\sigma_K : K \hookrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \simeq \mathbb{R}^n,$$

where $r_1 + 2r_2 = n$.

For $x \in K$,

$$\sigma_K(x) = (\sigma_1(x), \dots, \sigma_{r_1}(x), \Re(\sigma_{r_1+1}(x)), \Im(\sigma_{r_1+1}(x)), \dots, \Re(\sigma_{r_2}(x)), \Im(\sigma_{r_2}(x))).$$

Algebraic lattice

If $M \subset K$ is a free $\mathbb{Z}$ -module of rank n , then $\sigma_K(M) \subset \mathbb{R}^n$ is a lattice. In particular,

$$\Lambda_K := \sigma_K(\mathcal{O}_K)$$

is the algebraic lattice associated with the ring of integers.

Construction A

Let K be a number field, let $\mathfrak{p} \subset \mathcal{O}_K$ be a prime ideal, and assume $\mathcal{O}_K/\mathfrak{p} \cong \mathbb{F}_q$.
If $\mathcal{C} \subset \mathbb{F}_q^n$ is a linear code, define the reduction map $\rho : \mathcal{O}_K^n \longrightarrow \mathbb{F}_q^n$.

Construction A, (Conway and Sloane 2013; Costa et al. 2017)

The lattice is the preimage of the code:

$$\Lambda_A(\mathcal{C}) = \rho^{-1}(\mathcal{C}) = \{x \in \mathcal{O}_K^n : \rho(x) \in \mathcal{C}\}.$$

So Construction A lifts a linear code over a finite field to a lattice in Euclidean space via the canonical embedding.

$$\mathfrak{p}\mathcal{O}_K^n \subset \Lambda_A(\mathcal{C}) \subset \mathcal{O}_K^n.$$

Suppose that $\mathfrak{P} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$, where $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ are pairwise coprime prime ideals. By the Chinese Remainder Theorem,

$$\mathcal{O}_K/\mathfrak{P} \cong \mathcal{O}_K/\mathfrak{p}_1 \times \cdots \times \mathcal{O}_K/\mathfrak{p}_r.$$

Definition 2 (Construction π_A lattice, [Huang and Narayanan, *IEEE TIT*, 2017])

Let $\mathcal{C}_j \subseteq (\mathcal{O}_K/\mathfrak{p}_j)^n$ be (n, k_j) -linear codes for $j = 1, \dots, r$, and define

$$\mathcal{C} := \phi^{-1}(\mathcal{C}_1 \times \cdots \times \mathcal{C}_r) \subseteq (\mathcal{O}_K/\mathfrak{P})^n.$$

The corresponding Construction π_A lattice is $\Lambda_{\pi_A}(\mathcal{C}) := \rho^{-1}(\mathcal{C}) \subseteq \mathcal{O}_K^n$. Equivalently, $\Lambda_{\pi_A}(\mathcal{C}) = \mathcal{C} + \mathfrak{P}^n$.

Binary example over $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$

In our running example, $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$, the prime 2 splits completely:

$$\langle 2 \rangle = \mathfrak{p}_1 \mathfrak{p}_2 \mathfrak{p}_3 \mathfrak{p}_4.$$

By the Chinese Remainder Theorem,

$$\begin{aligned} \phi : \mathcal{O}_K / \langle 2 \rangle &\cong \mathbb{F}_2 \times \mathbb{F}_2 \times \mathbb{F}_2 \times \mathbb{F}_2 \\ \alpha &\mapsto (\alpha \bmod \mathfrak{p}_1, \alpha \bmod \mathfrak{p}_2, \alpha \bmod \mathfrak{p}_3, \alpha \bmod \mathfrak{p}_4) \in (\mathbb{F}_2)^4. \end{aligned}$$

For codewords of length n , $(\mathcal{O}_K / \langle 2 \rangle)^n \cong (\mathbb{F}_2^n)^4$. Therefore, we use four binary component codes $\mathcal{C}_j \subseteq \mathbb{F}_2^n$, $j = 1, \dots, 4$, and combine them through Construction π_A .

Main consequence

Four binary levels enable multistage decoding with lower complexity.

Nested lattice codes from Construction π_A

Let $\mathfrak{P} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$, and for each level j , choose nested codes

$$\mathcal{C}_e^{(j)} \subseteq \mathcal{C}_b^{(j)} \subseteq (\mathcal{O}_K/\mathfrak{p}_j)^n.$$

Using the CRT, define

$$\mathcal{C}_e \subseteq \mathcal{C}_b \subseteq (\mathcal{O}_K/\mathfrak{P})^n.$$

Applying Construction π_A we get,

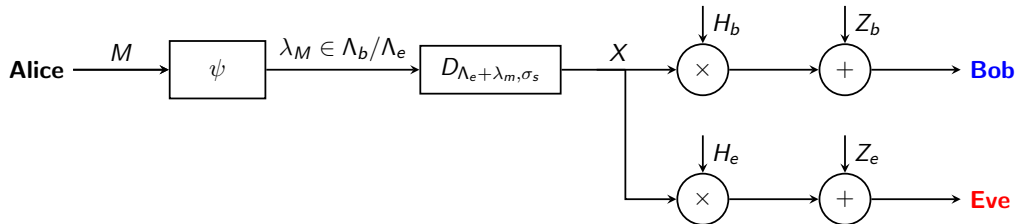
$$\tilde{\Lambda}_e = \mathcal{C}_e + \mathfrak{P}^n, \quad \tilde{\Lambda}_b = \mathcal{C}_b + \mathfrak{P}^n, \quad \tilde{\Lambda}_e \subseteq \tilde{\Lambda}_b.$$

After canonical embedding and scaling,

$$\Lambda_e = \gamma \sigma_K(\tilde{\Lambda}_e), \quad \Lambda_b = \gamma \sigma_K(\tilde{\Lambda}_b),$$

with $\Lambda_e \subseteq \Lambda_b \subset \mathbb{R}^{nm}$.

Wiretap channel



Alice sends a confidential message to Bob, while Eve also observes the transmission.

Goal

Design a coding scheme such that **Bob decodes reliably** and **Eve learns essentially nothing**.

In our approach, this is achieved by a nested lattice pair $\Lambda_e \subseteq \Lambda_b$.

Compound block-fading wiretap channel

We consider the complex block-fading wiretap channel [Campello, Ling et al., *IEEE TIT*, 2020]

$$Y_b = H_b X + Z_b, \quad Y_e = H_e X + Z_e,$$

where $H_b \in \mathbb{C}^{n_b \times n_a}$, $H_e \in \mathbb{C}^{n_e \times n_a}$, and Z_b, Z_e are Gaussian noise matrices.

Compound model

The exact channel matrices are not known. We only know that

$$H_b \in S_b, \quad H_e \in S_e.$$

The compound sets are defined by

$$S_b = \left\{ H : \left| I_{n_a} + \text{SNR}_b H^\dagger H \right| \geq e^{C_b} \right\}, \quad S_e = \left\{ H : \left| I_{n_a} + \text{SNR}_e H^\dagger H \right| \leq e^{C_e} \right\}.$$

We want lattice code constructed from the fine lattice Λ_b to be reliably decoded for every admissible Bob channel

$$H_b \in S_b.$$

Universal reliability, [Campello, Ling et al., *IEEE TIT*, 2018]

Find a sequence of fine lattices $\{\Lambda_b^{(n)}\}$ such that

$$P_e(H_b) \rightarrow 0 \quad \text{uniformly for all } H_b \in S_b.$$

Two ingredients are needed: Minkowski–Hlawka averaging and compactification.

Let $\mathbb{L}_n$ denote the ensemble obtained by choosing the component codes independently from balanced sets of codes and using Construction π_A to construct the lattices.

Minkowski–Hlawka theorem [S. et al., *IEEE TIT*, 2025]

For the ensemble $\mathbb{L}_n$ described above, for every non-negative continuous f with compact support,

$$\mathbb{E}_{\Lambda_b \in \mathbb{L}_n} \left[\sum_{\lambda \in \Lambda_b \setminus \{0\}} f(\lambda) \right] \leq \frac{1}{\text{vol}(\Lambda_b)} \int_{\mathbb{R}^N} f(\mathbf{u}) d\mathbf{u} + \varepsilon_n,$$

with $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$

- It shows that, on average, the lattice-point distribution of the ensemble behaves like that of a random lattice.
- Hence one can extract a sequence of fine lattices with the geometric goodness required for reliable decoding at Bob.

Compactification

For every normalized channel matrix $\tilde{H}$ with $\det(\tilde{H}^\dagger \tilde{H}) = 1$, one can write $\tilde{H} = E_{\tilde{H}} U_{\tilde{H}}$, where $\|E_{\tilde{H}}\|$ is uniformly bounded and $U_{\tilde{H}}$ preserves the distribution of the ensemble.

- It reduces the whole normalized compound channel family to a bounded set of equivalent channels.

This reduction uses the unit structure of $\mathcal{O}_K$ through Dirichlet's unit theorem.

Corollary 1 (Universal reliability at Bob)

There exists a sequence of lattices $\Lambda_b^{(n)} \in \mathbb{L}_n$ that is universally good for Bob's compound channel S_b , in the sense of [Campello, Ling et al., IEEE TIT, 2018].

Our secrecy goal is

$$\sup_{H_e \in S_e} I(M; Y_e) \rightarrow 0,$$

i.e., Eve's observation should be essentially independent of the message, [Ling et al., *IEEE TIT*, 2014].

Discrete Gaussian shaping

For each message $s \in M_n$, Alice samples a point in the coset $\Lambda_e + \lambda_s$ according to the discrete Gaussian distribution

$$D_{\Lambda_e + \lambda_s, \sigma_s}.$$

If there exist a reference density q_{Y_e} and $\varepsilon_n \in [0, 1/4]$ such that

$$\mathbb{V}(p_{Y_e|M=s}, q_{Y_e}) \leq \varepsilon_n \quad \text{for all } s \in M_n,$$

then $I(M; Y_e) \rightarrow 0$ whenever $\varepsilon_n = o(1/n)$.

For a fixed $H_e \in S_e$, the variational-distance parameter is bounded by

$$\varepsilon_n \leq \epsilon_{H_e \Lambda_e}(\sqrt{\Sigma_3}), \quad \Sigma_3^{-1} = (H_e H_e^\dagger)^{-1} \sigma_s^{-2} + \sigma_e^{-2} I.$$

Algebraic reduction

Assume that the dual lattice Λ^* admits algebraic reduction with constant $\alpha > 0$. Then, for every $H_e \in S_e$,

$$\epsilon_{H_e \Lambda_e}(\sqrt{\Sigma_3}) \leq \epsilon_{\Lambda_e}(\sigma_{\text{eq}}), \quad \sigma_{\text{eq}}^2 = \sigma_s^2 \alpha^{-2} e^{-C_e/n_a},$$

where $C_e := \sup_{H_e \in S_e} \log \det \left(I + \frac{\sigma_s^2}{\sigma_e^2} H_e H_e^\dagger \right)$.

In our construction, the coarse lattice Λ_e is a Construction π_A lattice built from Chinese remainder codes from a balanced set.

Minkowski–Hlawka

The corresponding ensemble satisfies the Minkowski–Hlawka property, so we can choose a sequence $\{\Lambda_e^{(n)}\}$ such that

$$\epsilon_{\Lambda_e^{(n)}}(\sigma_{\text{eq}}) = o(1/n).$$

Therefore,

$$\sup_{H_e \in S_e} \epsilon_{H_e \Lambda_e^{(n)}}(\sqrt{\Sigma_3}) \leq \epsilon_{\Lambda_e^{(n)}}(\sigma_{\text{eq}}) = o(1/n),$$

and hence

$$\sup_{H_e \in S_e} I(M; Y_e) \rightarrow 0.$$

Main theorem

Let $\{(\Lambda_b^{(n)}, \Lambda_e^{(n)})\}_{n \geq 1}$ be nested lattices from multiquadratic Construction π_A , with $\Lambda_e^{(n)} \subset \Lambda_b^{(n)}$. If

- $\Lambda_b^{(n)}$ is universally good for S_b , and
- $\Lambda_e^{(n)}$ is secrecy-good at σ_{eq} , with dual admitting algebraic reduction,

then there exists a discrete Gaussian encoder and a lattice decoder such that, for every $(H_b, H_e) \in S_b \times S_e$, Bob's error probability and the leakage $I(M; Y_e)$ both vanish as $n \rightarrow \infty$.

The achievable secrecy rate is

$$R < (C_b - C_e - n_a - 2n_a \log \alpha)^+.$$

Conclusion

We presented Construction π_A over multiquadratic fields as a framework for building multilevel lattice codes over $\mathcal{O}_K$. Completely split primes provide small residue alphabets, in particular binary levels, which make multistage decoding more practical. We also prove that there exist a sequence of nested lattices built from Construction π_A that achieve universal reliability and strong secrecy for the compound block-fading wiretap channel.

Perspectives

- Level-wise secrecy rate allocation, [Campello, Liu et al. 2017], [Luzzi et al., *IEEE TIT*, 2018].
- Finite-blocklength analysis for short-packet scenarios, [Polyanskiy et al., *IEEE TIT*, 2010]; [Hayashi, *IEEE TIT*, 2006]; [Bloch and Laneman, *IEEE TIT*, 2013].

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Thank you for your attention!

